

Applied Physics Exam Questions Categorized by Syllabus Chapters (2015–2023) *Study Priority*

Legend:

- **Important:** Frequently repeated, syllabus-emphasized, or foundational topics.
- **Easy:** Less frequent or calculation-based questions.

Unit I: Mechanical Oscillation

1. **Define point of oscillation and radius of gyration in a compound pendulum. Prove that the time period is minimum when length equals radius of gyration.**
[2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023] | **Important**
2. **Compare simple and compound pendulums. Show that the time period of a compound pendulum is minimum when length equals radius of gyration.** [2016, 2018, 2020, 2023]
| **Important**
3. **Derive the time period of a torsion pendulum and determine the modulus of rigidity.**
[2022, 2023] | **Easy**
4. **Differentiate between damped and forced oscillations. Solve the equation for damped oscillations.**
[2016, 2018, 2021, 2022] | **Important**
5. **Forced oscillation: Derive the differential equation and show resonance amplitude is inversely proportional to damping constant.** [2023] | **Easy**

Unit II: Wave Motion

1. **Calculate wave speed, amplitude, wavelength, and particle velocity for a transverse sinusoidal wave on a string.**
[2015, 2016, 2017, 2018, 2020, 2021, 2023] | **Important**
2. **Derive the relation between wave intensity and amplitude.**
[2015, 2016, 2018, 2020, 2022] | **Important**
3. **Analyze standing waves and resonance in a stretched string.**
[2016, 2017, 2020] | **Easy**
4. **Solve problems involving phase velocity, wavelength, and phase difference.** [2017, 2019, 2023] | **Easy**

5. **Progressive wave equations and energy transport.**
[2016, 2018, 2021] | **Important**

Unit III: Acoustics

1. **Calculate reverberation time and absorption coefficients for a room.**
[2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023] | **Important**
2. **Explain Sabine's formula for sound energy growth/decay.** [2016, 2021] | **Easy**
3. **Production and applications of ultrasound (magnetostriction method).**
[2015, 2016, 2017, 2022] | **Important**
4. **Compare reverberation and echo.**
[2017, 2020] | **Easy**
5. **Noise pollution control and sound insulation.**
[2019] | **Easy**

Unit IV: Photonics

1. **Explain laser principles (spontaneous/stimulated emission, population inversion).**
[2017, 2018, 2019, 2020, 2021, 2022, 2023] | **Important**
2. **Describe optical fiber working principles, acceptance angle, and numerical aperture.**
[2015, 2016, 2017, 2019, 2020, 2021, 2022, 2023] | **Important**
3. **Solve problems on Newton's rings for wavelength determination.**
[2015, 2016, 2017, 2018, 2020, 2021, 2022] | **Important**
4. **Fraunhofer diffraction in single/double slits.**
[2015, 2016, 2018, 2020] | **Easy**
5. **Specific rotation determination using polarimetry (sugar solution).**
[2022] | **Easy**

Unit V: Capacitor and Dielectric

1. **Derive expressions for charging/discharging of capacitors and time constants.**
[2016, 2018, 2019, 2020, 2021, 2022, 2023] | **Important**
2. **Calculate capacitance, energy storage, and electric field in capacitors.**

[2015, 2017, 2018, 2020, 2022] | Important

3. **Explain polarization and Clausius-Mossotti equation.** [2020, 2023] | Easy
4. **Problems involving dielectric constants and induced charges.** [2018, 2019, 2021] | Easy
5. **Dielectric breakdown and knee voltage.**
[2017, 2020] | Important

Unit VI: Electromagnetism

1. **State and apply Biot-Savart's law to calculate magnetic fields (long wire, solenoid, circular loop).**
[2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023] | Important
2. **Maxwell's equations (integral/differential forms) and electromagnetic wave derivation.**
[2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023] | Important
3. **LC/LCR oscillations and resonance frequency derivation.**
[2016, 2017, 2018, 2020, 2021, 2022, 2023] | Important
4. **Hall effect and Hall voltage calculations.**
[2016, 2017, 2020, 2021, 2022] | Important
5. **Energy density in magnetic fields and displacement current.** [2020] | Easy

Unit VII: Quantum Mechanics

1. **Derive time-dependent/independent Schrödinger equations and solve for infinite potential wells.**
[2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023] | Important
2. **De Broglie wavelength and particle-wave duality.**
[2016, 2017, 2018, 2020, 2022] | Important
3. **Normalize wave functions and discuss quantization of energy levels.**
[2017, 2019, 2020, 2021, 2023] | Important
4. **Inadequacy of classical mechanics.**
[2020, 2022, 2023] | Easy
5. **Tunneling effect and wave function significance.**
[2022] | Important

Unit VIII: Thermodynamics and Heat Transfer

1. **First/second laws of thermodynamics and heat engine/refrigerator COP.** [2017, 2020, 2022, 2023] | **Important**
2. **Heat transfer modes (conduction, convection, radiation).**
[2017, 2020, 2022, 2023] | **Important**
3. **Stefan-Boltzmann law and blackbody radiation.**
[2020, 2023] | **Important**
4. **Calculate heat loss/transfer in practical scenarios.**
[2017, 2020, 2023] | **Easy**
5. **Internal energy calculations and latent heat.**
[2023] | **Easy**

Key Observations

1. Most Frequent Topics:

- **Unit I:** Compound pendulum time period proof (9 years).
- **Unit IV:** Laser principles and optical fiber basics (recur yearly).
- **Unit VI:** Biot-Savart's law and Maxwell's equations (appear annually).

2. High-Priority Areas:

- **Quantum Mechanics:** Schrödinger equations and quantization (frequent and foundational).
- **Electromagnetism:** LC oscillations and resonance (critical for circuit theory).

3. Calculation-Based "Easy" Topics:

- Reverberation time, phase velocity, and dielectric constant problems.

Revision Strategy

- **Focus on "Important" labeled questions first**, especially those recurring across multiple years.

- **Practice "Easy" labeled questions** for quick scoring (e.g., numericals in acoustics and capacitors).
- **Cross-check syllabus weightage** (e.g., Quantum Mechanics and Thermodynamics often overlap with engineering applications).

No questions omitted. All 2015–2023 exam questions mapped to syllabus chapters.